



LECTURE 1

DEVELOPMENT PROCESSES & ORGANIZATIONS

COEN / ELEC 390

Acknowledgement

- The content of these slides is based on the textbook :

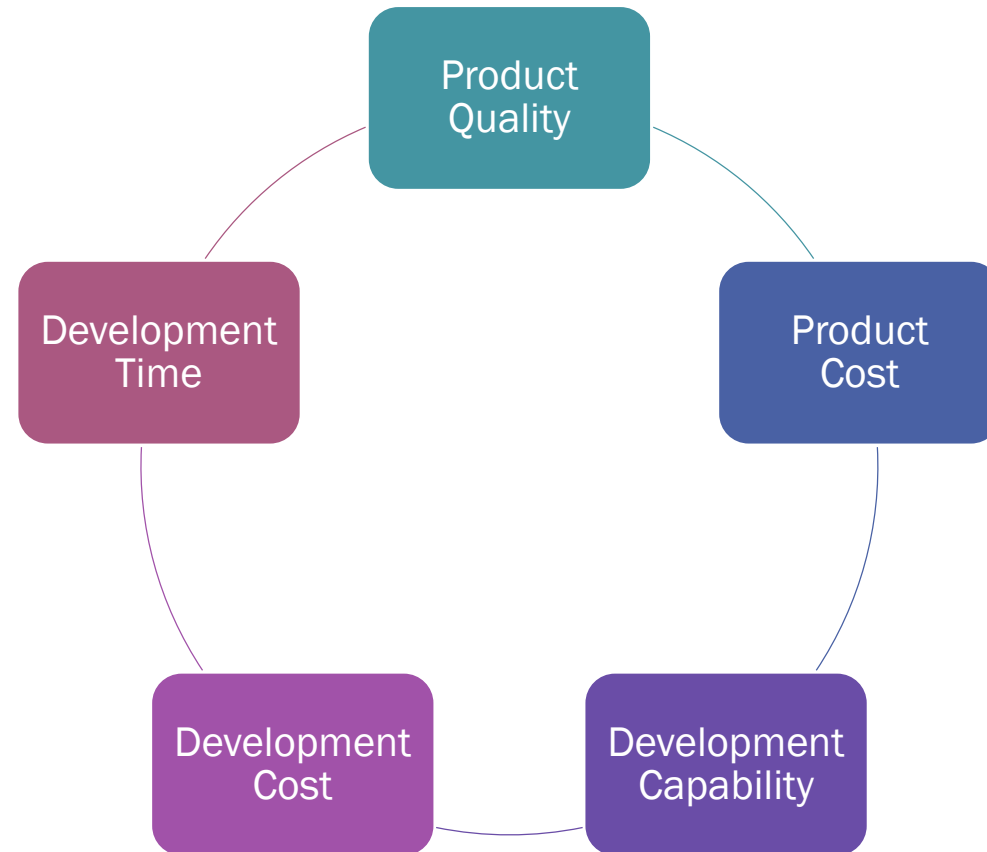
Product Design and Development by K.T. Ulrich, S.D. Eppinger
McGraw Hill Irwin

Product, Product Development, Process

- A **product** is something that is sold to customers.
- **Product development** refers to a set of activities that starts with a perception of a market opportunity and ends with delivery of a product.
- A **product development process** is a sequence of steps or activities which an enterprise employs to conceive, design, and commercialize a product.

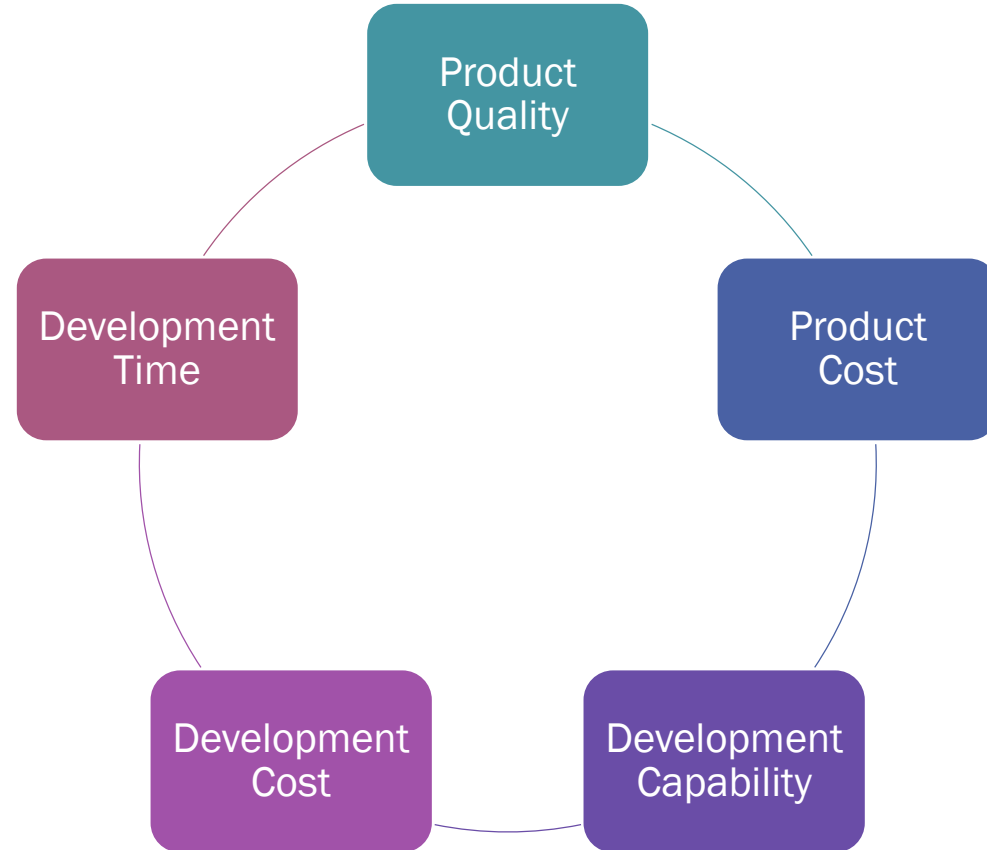
Characteristics of Product Development

- Product Quality
 - Does it satisfy customer needs? Is it reliable? Is it robust?
- Product Cost
 - Cost per unit (parts, labour, software licenses, other manufacturing costs)
 - Capital cost for per unit building (tooling, equipment)
 - Product cost is factor in unit price and potential profit



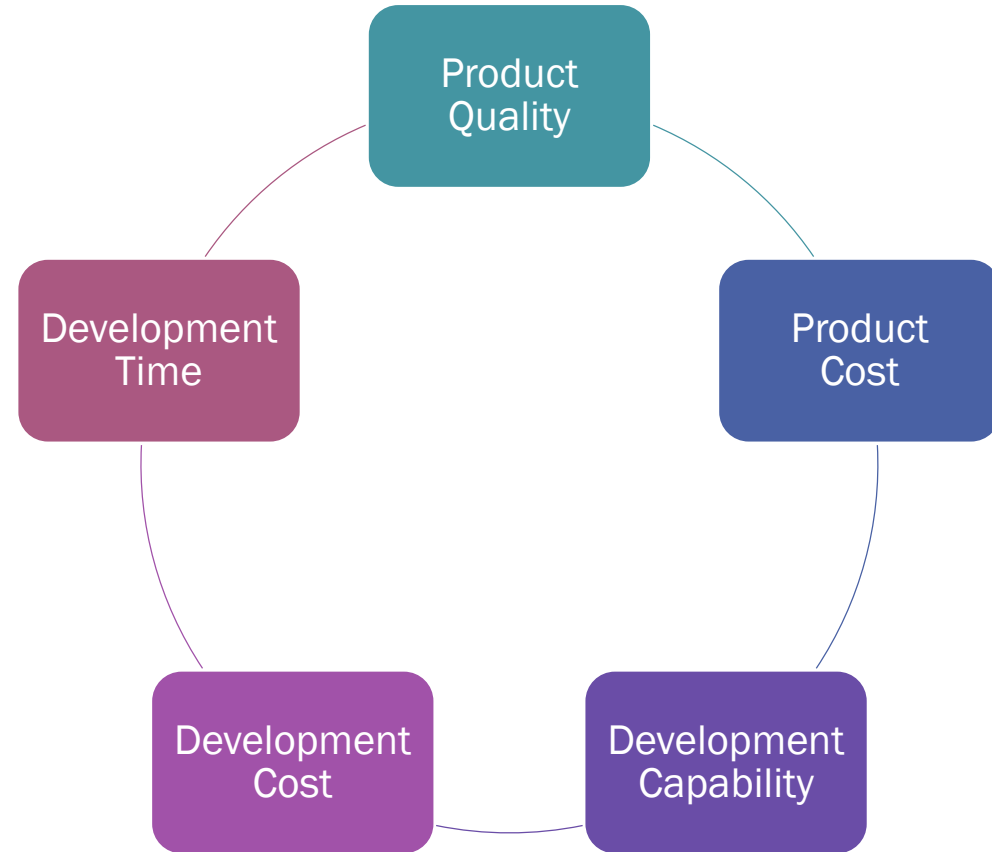
Characteristics of Product Development (Cont.)

- Development Time
 - How responsive to competition and to changes in technology?
- Development Cost
 - Personnel costs, facility costs, equipment costs, software licenses, etc.
- Development Capability
 - The firm has certain product development capabilities in terms of skilled employees and expertise.
 - will the capabilities be increased for use in future developments?



Tradeoffs

- These five characteristics are intertwined
- Examples:
 - Good initial development capability can reduce development time and development cost
 - Fast development time results in the same product having higher quality when it first comes out.



Actors in Design and Development

Marketing

- Help identify product opportunities, segmentation of the market, customer needs
- Communicates with customers
- Sets prices, launches and promotes product

Design

- Lead role in creating the product that meets the customers' needs
- Includes engineering design
- Industrial design (aesthetics/user interface)

Manufacturing

- Design/operate/co-ordinate production system
- Purchasing of components
- Distribution
- Installation
- Supply chain

Development and Production Costs (Ulrich & Eppinger)

	Stanly Tools Jobmaster Screwdriver	Rollerblade in- line skate	Hewlett Packard Deskjet Printer	Volkswagen new Beetle	Boeing 777 airplane
Volume	100,000/yr	100,000/yr	4 million/yr	100,000/yr	50/yr
Sales Life	40 years	3 years	2 years	6 years	30 years
Price	\$5	\$150	\$130	\$20,000	\$260 million
Dev. Time	1 year	2 years	1.5 years	3.5 years	4.5 years
Int Dv team	3	5	100	800	8,600
Ex Dv team	3	10	75	800	10,000
Dev Cost	\$150K	\$750K	\$50 M	\$400 M	\$3 B
Prod Invest	\$150K	\$1M	\$25M	\$500M	\$3 B

Challenges in Product Development

- **Trade-offs:** recognize, understand, manage to maximize success of the product.
- **Dynamics:** Constant changes (new technology, new customer preferences, new products from competitors).
- **Details:** small decisions can have huge impact.
- **Time pressure:** decisions must be taken quickly, with incomplete information.
- **Economics:** large amounts of investments are at stake

Product Development Processes

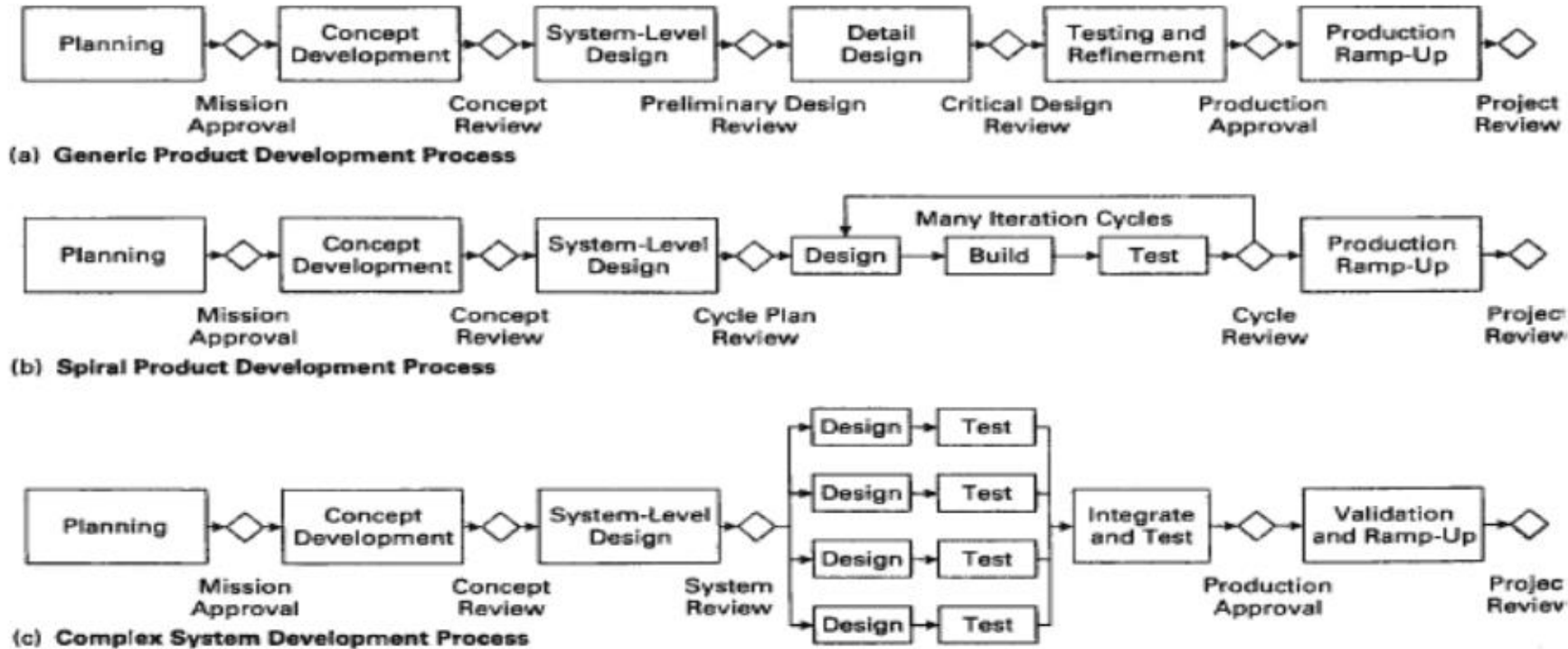


EXHIBIT 2-5 Process flow diagrams for three product development processes.

Other Ways to Look at Product Development

- **Option Narrowing:** The initial creation of a wide set of alternative product concepts followed by narrowing of alternatives and increasing specification of the product until the product can be reliably produced.
- **Information Gathering:** The process begins with inputs such as the corporate objectives and concludes when all the information required to support production & sales are created and communicated.
- **Risk Reduction:** In the early phases of development, various risks are identified and prioritized. As the progress is made, risks are reduced.

Product Types

- **Market Pull products**

- Start with market opportunity
- Select appropriate technologies to meet customer needs

- **Technology-Push Products**

- Start with new technology
- Find a market for the new technology
- Unlikely to succeed unless the new technology offers clear competitive advantage

- **Platform Products**

- Built around a pre-existing system
- A large investment is made to create new platforms.
- Example: Smartphones

- **Process Intensive Products**

- Products that are highly dependent on the production process
- Product design cannot be separated from the production process development
- Examples: food, semiconductors, chemicals, steel, paper.

Product Types (Cont.)

- **Customized Products**

- New products are slight variation of existing products
- Often involves setting design variables and then following a well structured process to create new product
- Examples: motors, batteries, switches, containers

- **High-Risk Products**

- Elevated risks technology, market, budget, etc.
- Multiple solution paths are explored to try to ensure that at least one succeeds.
- Design reviews must check to make sure risks are being reduced over time, not just postponed.
- Examples: pharmaceuticals, space systems

Product Types (Cont.)

- **Quick-build Products**

- Build and test prototypes are very rapid. Thus this part of the product design process can be done repeatedly and quickly.
- This is sometimes called the spiral product development process.
- This process takes advantage of the fast prototyping cycle by using the result of each cycle to learn to how modify the next cycle. Customers are involved in testing after one or more cycles.
- Often the build/test cycle is repeated until time or budget runs out
- Examples: software, electronics.

- **Complex Systems**

- The product is made of many interacting subsystems.
- System level design, via decomposition, is critical.
- Detailed design of components is a highly parallel process in which many development teams work at the same time, separately.
- The testing and refinement phase includes system integration as well as extensive testing and validation at every level.
- Examples: airplanes, automobiles